Project Check-in 1: Park Check, a Grounded Trip Planner for California National Parks

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Project Activities Completed:

- Collected 34 sources on RAG, tool use, hallucination and prompt injection
- Built a cached NPS Data API client and five tools (park info, alerts, activities, campgrounds, fees), with park codes limited to the 12 supported parks.
- Built a tool calling loop and three configs: A (gpt-5.6-terra with tools), B (same model, no tools) and C (gpt-5.6-luna with tools).
- Wrote 24 test queries and ran the first live test. It found a bug where the models rejected tools unless the reasoning effort was "none", which I fixed for all configs.
- Ran all 72 query and config pairs on one saved API snapshot (0 errors, $0.32). Park recall was 0.93 for A and 1.00 for C, and C was about 7 times cheaper than A. Without tools, B guessed from memory, for example saying Tioga Road is "typically closed by mid-October".

Project Activities in Progress:

- Claim level scoring: an LLM judge splits each answer into claims and checks each one against the API snapshot, so I can measure accuracy and not just tool use.

Project Activities Planned

- Six safety tests, a Streamlit demo app, peer testing with two classmates, and the README, slides and demo video.

GitHub/Google Drive links:

https://github.com/SShahparnia/cmpe259_project

AI Disclosure

I used Claude Code to help write boilerplate code and debug API issues (which I ran into many of). I ran the evaluation reported here myself and checked/interpreted the results.

References:

I didn't use many of these papers. They were just to help me learn more about the topics for the project. I've done projects on LLM as a judge and RAG before so many are repeat papers that I had already read in the past.

- Lewis et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. https://arxiv.org/abs/2005.11401
- Gao et al. (2023). Retrieval-Augmented Generation for Large Language Models: A Survey. https://arxiv.org/abs/2312.10997
- Kasai et al. (2023). RealTime QA: What's the Answer Right Now? https://arxiv.org/abs/2207.13332
- Yao et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. https://arxiv.org/abs/2210.03629
- Patil et al. (2023). Gorilla: Large Language Model Connected with Massive APIs. https://arxiv.org/abs/2305.15334
- Li et al. (2023). API-Bank: A Comprehensive Benchmark for Tool-Augmented LLMs. https://arxiv.org/abs/2304.08244
- Qin et al. (2024). ToolLLM: Facilitating Large Language Models to Master 16000+ Real-world APIs. https://arxiv.org/abs/2307.16789
- Xie et al. (2024). TravelPlanner: A Benchmark for Real-World Planning with Language Agents. https://arxiv.org/abs/2402.01622
- Wei et al. (2024). QCG-Rerank: Chunks Graph Rerank with Query Expansion in Retrieval-Augmented LLMs for Tourism Domain. https://arxiv.org/abs/2411.08724
- Ji et al. (2023). Survey of Hallucination in Natural Language Generation. https://arxiv.org/abs/2202.03629
- Chen et al. (2024). Benchmarking Large Language Models in Retrieval-Augmented Generation. https://arxiv.org/abs/2309.01431
- Chen et al. (2023). FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance. https://arxiv.org/abs/2305.05176